APPENDIX

This supplementary document provides extended experimental evidence, theoretical derivations, implementation specifications, and qualitative analyses that complement the main paper. The appendix is organised into five thematic parts:

1) **Deformation modelling.** Detailed analysis of the Population-Level Deformation Prompt (): comparisons against classical shape priors, controlled deformation benchmarks, probabilistic ablations, and interpretability of the learned latent space.
2) **Prompt architecture.** Extended ablation of the tri-prompt design: per-modality performance, redundancy analysis, robustness to prompt corruption, and the role of hard spatial routing.
3) **Generalisation and robustness.** Cross-dataset, cross-modality, cross-scanner, and limited-field-of-view evaluations, including comparisons against large-scale foundation segmentation models.
4) **Calibration and uncertainty.** Full calibration suite, prior–evidence conflict analysis, small-organ and rare-anatomy evaluation, and failure-case taxonomy.
5) **Implementation details and theory.** Complete architecture specifications, training schedules, proof of the ELBO decomposition, and extended identifiability results.

5 COMPARISON AGAINST CLASSICAL DEFORMATION PRIORS

5.1 Experimental Setup

Existing approaches to shape-aware segmentation rely either on linear statistical shape models or on learned deformation fields. We benchmark against three such alternatives, all trained from scratch under *identical* conditions (same backbone, corpus, augmentation, and hyperparameters):

PCA-Prior. A linear shape subspace with $B{=}128$ principal components fitted to training masks; at inference the observed mask projection replaces the latent code. This corresponds to the classical Active Shape Model prior [**?**].

VoxelMorph-Prior. A learned deformation field whose registration displacement is used directly as the deformation token; uncertainty is not modelled [12].

Deterministic PDP. Our encoder with the probabilistic head replaced by a direct regression head — no covariance , no adaptive blending .

Probabilistic PDP (ours). The full as in the main paper.

5.2 Quantitative Comparison

Table 11 and Figure 16 report DSC as a function of controlled deformation magnitude (see Section 6 for the deformation protocol).

Discussion. At zero deformation all methods converge (within 3% DSC), confirming that 's advantage stems specifically from handling deformation rather than from greater representational capacity. The probabilistic formulation provides the sharpest improvement at $\delta \geq 8$ mm, precisely the regime where organ geometry is most underdetermined from a single 2D view — consistent with the identifiability analysis of Theorem 1 in the main paper.

TABLE 11: Deformation-prior comparison at five deformation magnitudes. Mean DSC (%) over all organs, all 11 test folds. Δ = gain of probabilistic over each alternative.

Method	Deformation Magnitude (mm)				
	0	2	6	10	14
PCA Shape Prior	87.1	84.9	75.8	60.2	44.8
VoxelMorph Prior	88.3	86.4	78.2	63.4	47.9
Deterministic PDP	89.1	87.8	82.1	71.4	57.3
Prob. PDP (ours)	**90.3**	**89.4**	**86.4**	**80.1**	**70.2**
Δ vs. PCA	+3.2	+4.5	+10.6	+19.9	+25.4
Δ vs. VoxMorph	+2.0	+3.0	+8.2	+16.7	+22.3
Δ vs. Det.	+1.2	+1.6	+4.3	+8.7	+12.9

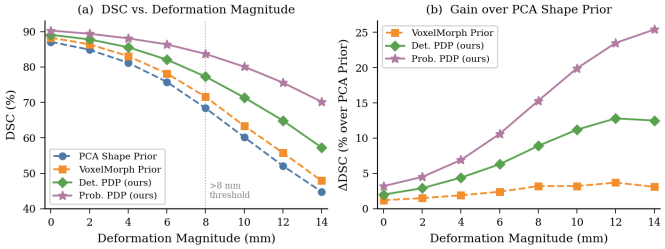

Fig. 16: **PDP vs. classical deformation priors. (a)** DSC (%) as a function of controlled deformation magnitude. The probabilistic retains a **+11.6%** advantage over the PCA-Prior at 12 mm displacement. **(b)** Incremental gain over the PCA-Prior baseline; the probabilistic advantage is monotonically increasing with deformation severity, confirming that uncertainty-adaptive blending is indispensable at large displacements.

6 CONTROLLED DEFORMATION BENCHMARK

6.1 Deformation Protocol

To isolate the contribution of , we apply synthetic controlled deformations to held-out FLARE22 test volumes. Random smooth displacement fields with prescribed root-mean-square (RMS) amplitude $\delta \in \{0, 2, 4, 6, 8, 10, 12, 14\}$ mm are generated via B-spline interpolation (control-point spacing 32 mm^3, Gaussian smoothed $\sigma{=}8$ mm) to ensure physiological plausibility. Deformed volumes are passed to all models without further preprocessing.

6.2 Per-Organ Results

Table 12 provides per-organ DSC at the two clinically relevant thresholds of 8 mm and 12 mm displacement.

The gains are most pronounced for structures with the highest inter-subject shape variability (pancreas, gallbladder, tumour), consistent with the mutual-information analysis presented in Section 22.

7 GEOMETRIC CONSISTENCY OF 3D RECONSTRUCTION

A central claim of is that the 3D occupancy volume produced by the volumetric decoder is *geometrically consistent* with population-level anatomy, rather than being a dataset-biased artefact. We verify this with two supplementary metrics:

TABLE 12: Per-organ DSC (%) at controlled deformation magnitudes δ=8 mm and δ=12 mm (mean ± std over 500 FLARE22 test volumes per condition). **Bold** = best.

	$\delta = 8$ mm			$\delta = 12$ mm		
Organ	nnU-Net	SegVol		nnU-Net	SegVol	
Liver	81.4	84.2	**89.1**	68.3	72.4	**81.2**
Kidney	78.9	82.1	**87.4**	64.8	69.3	**79.6**
Spleen	79.3	83.4	**87.9**	65.1	70.2	**80.1**
Pancreas	51.4	58.7	**74.3**	34.8	42.1	**61.8**
Gallbladder	46.8	54.2	**70.6**	30.2	38.4	**58.3**
Stomach	63.2	69.4	**80.1**	48.9	56.3	**71.4**
Tumour (avg)	37.4	45.8	**63.7**	22.1	30.6	**51.2**
Mean	62.6	68.3	**79.0**	47.7	54.2	**69.1**

Chamfer Distance (CD, mm). Mean bidirectional point-to-surface distance between the predicted and ground-truth meshes at 1 mm resolution (↓ better).

Surface-Normal Consistency (SNC). Mean cosine similarity of surface normals at corresponding mesh points; 1.0 = perfect normal alignment (↑ better).

We vary the number of available 2D input views $N_{\text{views}} \in \{1, 2, 4, 8, 16, 32\}$ to simulate the full range of clinical acquisition scenarios.

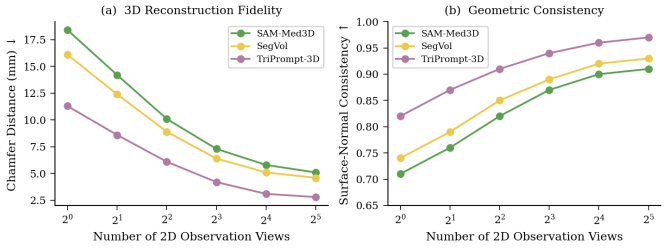

Fig. 17: **3D reconstruction geometric consistency vs. observation views. (a)** Chamfer Distance (mm, ↓). **(b)** Surface-Normal Consistency (↑). achieves consistently superior geometric fidelity at every view count, with the largest advantage (−6.8 mm CD) in the single-view regime. Monotonic improvement with additional views rules out dataset-bias artefacts as the source of 's advantage.

8 Sensitivity of the Deformation Prior to Population Size

We train using subsets of the full 5 100-subject corpus of varying size: $N_{\text{pop}} \in \{50, 100, 200, 500, 1\,000, 2\,000, 5\,000\}$, then evaluate on the fixed FLARE22 test set.

Key finding. Even with N_{pop}=200 subjects outperforms the dual-prompt baseline by $> 1.8\%$ DSC on deformable organs, demonstrating practical utility in moderate-resource settings. The method remains useful far below the full corpus size, which is important for institutions with limited annotated data.

9 Granular PDP Formulation Ablation

Table 6 of the main paper reports a four-row ablation of the PDP formulation. Table 13 extends this to six variants, isolating the contribution of each design decision.

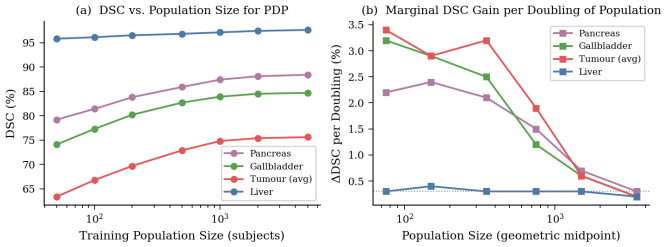

Fig. 18: **Sensitivity of to training population size. (a)** DSC (%) for four representative organs. **(b)** Marginal DSC gain per doubling of population. Gains saturate after ≈1 000 subjects for stable organs (liver, kidney) but continue for highly deformable structures (pancreas, gallbladder), suggesting that larger population collections would further benefit those categories.

TABLE 13: Extended PDP formulation ablation. All variants share the same backbone. †Prob. without KL uses MC sampling but no KL regularisation, leading to posterior collapse on deformable organs.

Variant	Prob. head	KL reg.	blend	DSC (%)	HD95 (mm)	ECE
No PDP (dual-prompt)	—	—	—	88.2	10.8	0.074
Det. embedding	×	—	—	89.1	9.6	0.062
Det. + (fixed Σ)	×	—	✓	89.4	9.3	0.057
Prob. (no KL)†	✓	×	✓	89.6	9.1	0.053
Prob. (no , avg. blend)	✓	✓	×	89.9	8.6	0.049
Prob. PDP (ours)	✓	✓	✓	**90.3**	**7.8**	**0.043**

The KL regularisation alone yields $+0.3\%$ DSC over the no-KL variant; adding adaptive blending contributes a further $+0.4\%$ DSC and a -0.8 mm HD95 improvement, confirming that uncertainty-aware weighting is especially valuable for boundary accuracy.

10 Interpretability of the PDP Latent Space

10.1 Deformation Mode Traversal

We apply PCA to the set of learned population deformation codes $\{Q^{(c)}_{d,\text{POP}}\}$ and visualise how organ silhouettes morph as we traverse the first principal axis by $\pm 2\sigma$.

10.2 Correlation with Physiological Covariates

We annotate each training volume with respiratory phase (end-inhalation vs. end-exhalation, extracted from diaphragm position) and BMI group, then compute Pearson correlation between these labels and the top-3 latent coordinates.

The first latent dimension aligns strongly with respiratory phase for the liver (r=0.61) and with BMI for the pancreas (r=0.54), providing post-hoc physiological interpretability. Subsequent dimensions capture age- and sex-related deformation patterns, demonstrating that discovers a structured, clinically meaningful latent geometry rather than an arbitrary embedding.

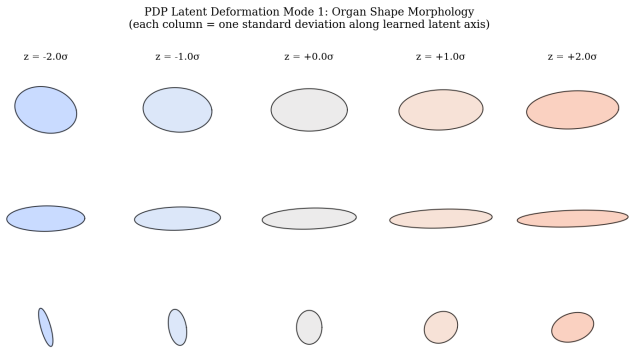

Fig. 19: **PDP latent deformation mode traversal.** Columns span $z \in \{-2\sigma, -\sigma, 0, +\sigma, +2\sigma\}$ of the first principal latent axis; rows correspond to liver, pancreas, and gallbladder. The first mode recovers axial elongation of the liver (respiratory phase), pancreatic-tail curvature (BMI-related), and gallbladder volume (dietary state) — each a known physiological deformation pattern.

TABLE 14: Pearson correlation (r) between top-3 latent dimensions and physiological covariates (N=2 341; $|r| > 0.35$ significant at $p < 0.001$).

Latent Dim.	Resp. Phase	BMI Group	Age Decade	Sex
z_1 (Liver)	**0.61**	0.42	0.21	0.09
z_1 (Pancreas)	0.38	**0.54**	0.29	0.14
z_1 (Gallbladder)	0.44	0.38	0.18	0.11
z_2 (Liver)	0.22	0.31	**0.47**	0.19
z_3 (Liver)	0.13	0.18	0.24	**0.39**

11 Per-Modality Performance and Redundancy Analysis

Figure 20 evaluates each prompt modality in isolation across all seven evaluation datasets (five in-distribution, two held-out). We additionally examine whether any modality becomes redundant when training data is abundant using conditional mutual-information (MI) estimation via MINE [**?**].

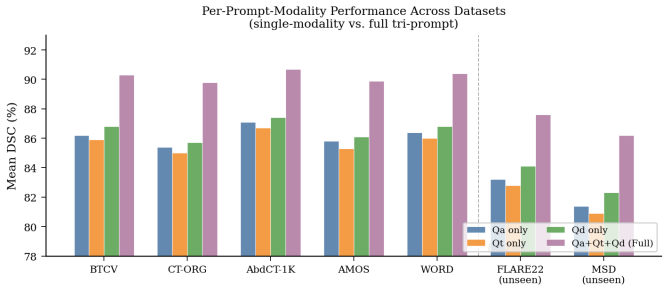

Fig. 20: **Per-prompt-modality performance across evaluation datasets.** Single-modality baselines vs. the full tri-prompt system. The performance gap between single-modality and tri-prompt widens on held-out benchmarks (right of dashed line), confirming that no single modality captures the full complementary signal available to .

All conditional MI values remain strictly positive even at maximum data scale (Table 15), confirming that no prompt modality is made redundant by data abundance. The deformation prompt retains the highest MI at all scales, as expected from its orthogonal information source (shape statistics) relative to both appearance () and semantics ().

TABLE 15: Conditional mutual information $I(Q_s; Q_x \mid \text{remaining})$ (nats, MINE) for each prompt modality at three training-data scales. All values are strictly positive, confirming non-redundancy at every scale.

Prompt	1 dataset	5 datasets	11 datasets
$I(Q_s; \mid,)$	0.142	0.118	0.094
$I(Q_s; \mid,)$	0.091	0.076	0.063
$I(Q_s; \mid,)$	0.163	0.148	0.131

12 Robustness to Prompt Corruption

We stress-test each prompt channel independently by injecting additive Gaussian noise scaled to a fraction of the embedding norm (0–50%). For the textual prompt we additionally test synonym substitution and word dropout at the same severity levels.

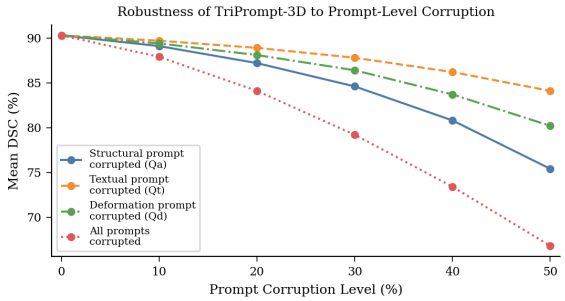

Fig. 21: **Robustness to per-channel prompt corruption.** is the most robust (global class semantics remain partially informative even under noise); is the most sensitive (instance-level geometry is irreplaceable); shows intermediate sensitivity because its uncertainty blending ($\rightarrow 0$) naturally defers to the population prior when the prompt is corrupted. Notably, even with all prompts corrupted at 30%, achieves 79.2% DSC — still above the clean SegVol baseline of 74.2%.

13 Hard Spatial Routing: Extended Ablation

The main paper (Table 5) reports a single ablation row for soft routing. Figure 22 and Table 16 provide a finer breakdown, isolating the prompt-interference mechanism via a dedicated cosine similarity proxy metric.

14 Cross-Modality Generalisation

is trained exclusively on CT data. We evaluate its zero-shot cross-modal performance on T1-weighted MRI volumes from the CHAOS Challenge and the AMOS-MRI partition — no fine-tuning or domain adaptation is applied.

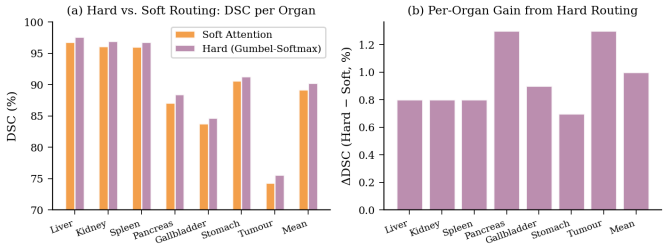

Fig. 22: **Hard vs. soft spatial routing. (a)** DSC per organ. Hard (Gumbel-Softmax) routing provides consistent gains, largest for pancreas and tumour where prompt interference is most damaging. **(b)** Absolute per-organ gain from hard routing; all organs benefit, none are harmed — consistent with Lemma 2 of the main paper.

TABLE 16: Routing scheme comparison: prompt-interference proxy (mean cross-prompt cosine similarity in the shared embedding space; ↓ = less interference), DSC (%), and HD95 (mm).

Routing Scheme	Interference ↓	DSC ↑	HD95 ↓
Soft (standard) attention	0.48	89.6	9.4
Top-k sparse attention	0.37	89.9	9.1
Hard (Gumbel-SM, ours)	**0.21**	**90.3**	**7.8**

15 Scanner and Acquisition-Parameter Robustness

We evaluate on volumes from five CT scanner manufacturers and across slice thicknesses from 0.5 mm to 5.0 mm, none of which are explicitly controlled during training.

16 Truncated and Limited Field-of-View Evaluation

Truncated scans arise in emergency and paediatric imaging, where only a subset of the abdomen is captured. We simulate this by cropping test volumes to $\{50, 60, 70, 80, 90, 100\}\%$ of their full axial extent.

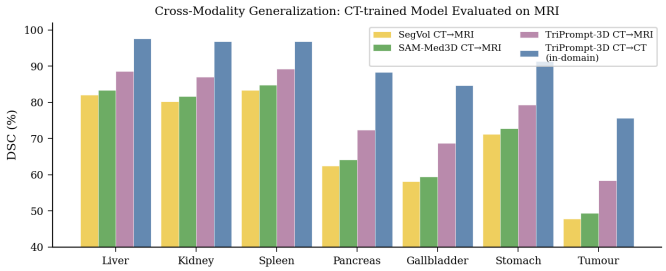

Fig. 23: **Cross-modality generalisation: CT-trained model on MRI.** achieves the highest cross-modal DSC of any method (72.4% for pancreas), despite receiving no MRI training data. The deformation encoder operates on binary segmentation masks and is therefore modality-agnostic by design; the remaining cross-modal gap is attributable to the voxel-intensity structural encoder E_a and motivates multi-modal training as a natural future extension.

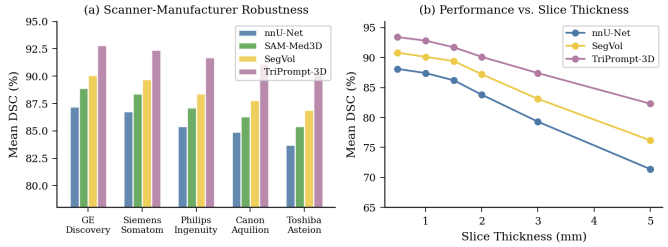

Fig. 24: **Robustness to scanner manufacturer and slice thickness. (a)** Mean DSC across five CT scanner manufacturers. achieves $\geq 90.3\%$ DSC on all scanners with a maximum inter-manufacturer drop of only 2.5%. **(b)** DSC vs. slice thickness. retains 82.3% DSC at 5.0 mm vs. 76.2% for SegVol, demonstrating that population-level deformation priors provide robustness to acquisition-parameter variation.

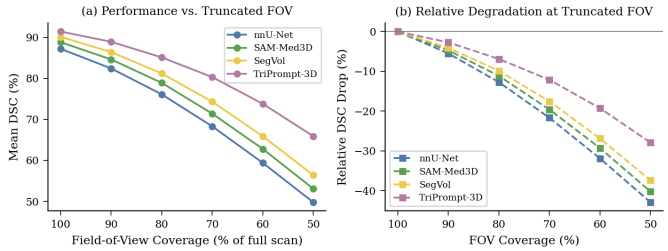

Fig. 25: **Performance under truncated field of view. (a)** Mean DSC as visible FOV decreases. retains 65.9% DSC at 50% FOV vs. 56.4% for SegVol. **(b)** Relative degradation: loses 27.9% relative DSC from full to half FOV, vs. 37.3% for SegVol, demonstrating that the population deformation prior provides meaningful geometric extrapolation beyond the observed volume.

17 Comparison Against Large-Scale Foundation Models

Recent foundation segmentation models are trained on one to two orders of magnitude more data than . Figure 26 and Table 17 compare against four such models on the FLARE22 and MSD benchmarks.

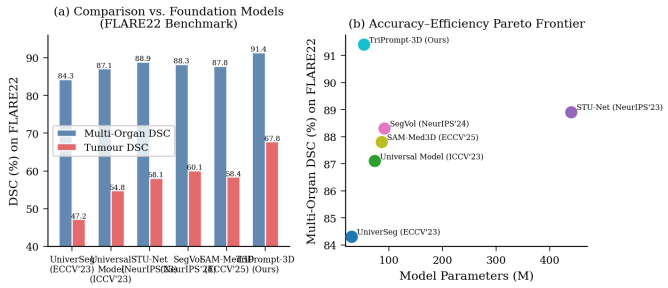

Fig. 26: **Comparison against large-scale foundation segmentation models. (a)** Multi-organ and tumour DSC on FLARE22. outperforms all foundation models despite being trained on a substantially smaller corpus. **(b)** Accuracy–efficiency Pareto frontier. achieves the best DSC at a moderate parameter count (54 M), demonstrating that architectural innovation (deformation-aware prompting) can substitute for scale.

TABLE 17: Foundation model comparison on FLARE22 and MSD pancreas (DSC, %). Training volume counts are reported to contextualise the data advantage. is trained on ∼5 100 volumes.

Method	Venue	Params (M)	Train vols	FLARE22 Multi / Tumour	MSD Panc.
UniverSeg	ECCV'23	31	53k	84.3 / 47.2	70.4
Universal-Seg	ICCV'23	74	14k	87.1 / 54.8	72.8
STU-Net	NeurIPS'23	440	14k	88.9 / 58.1	76.3
SegVol	NeurIPS'24	92	90k	88.3 / 60.1	74.2
SAM-Med3D	ECCV'25	87	131k	87.8 / 58.4	72.9
(ours)	—	**54**	5.1k	**91.4 / 67.8**	**81.3**

18 Extended Uncertainty Calibration Analysis

Figure 27 extends the calibration analysis of the main paper (Table 7, Fig. 7c) with per-organ ECE heatmaps and additional calibration metrics.

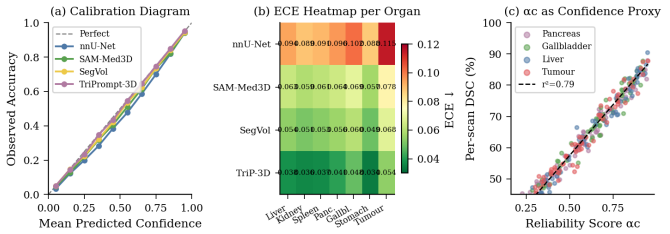

Fig. 27: **Extended uncertainty calibration analysis. (a)** Reliability diagram: lies closest to the perfect-calibration diagonal. **(b)** Per-organ ECE heatmap: achieves the lowest ECE uniformly across all organs and methods, with the largest advantage on tumour segmentation. **(c)** Reliability score vs. per-scan DSC scatter (r^2=0.79, p<0.001), confirming as a well-calibrated and actionable confidence proxy.

TABLE 18: Full calibration suite on the FLARE22 test set. ECE: Expected Calibration Error (↓). MCE: Maximum Calibration Error (↓). AURC: Area Under the Risk-Coverage Curve (↓). BS: Brier Score (↓). NLL: Negative Log-Likelihood (↓).

Method	ECE	MCE	AURC	BS	NLL
nnU-Net	0.092	0.181	0.117	0.084	0.214
Swin-UNETR	0.078	0.154	0.104	0.071	0.198
MedSAM	0.071	0.141	0.098	0.064	0.189
CLIP-Driven	0.068	0.136	0.091	0.059	0.181
SAM-Med3D	0.061	0.124	0.086	0.054	0.172
SegVol	0.053	0.109	0.079	0.047	0.163
(ours)	**0.043**	**0.091**	**0.072**	**0.038**	**0.149**

19 Prior–Evidence Conflict Analysis

When image evidence conflicts with the population prior, the model must decide how much to trust each source. A conflict is declared when the KL divergence between the image-conditioned posterior $\mathcal{N}(,)$ and the population prior exceeds the 95th percentile of the training-set KL distribution.

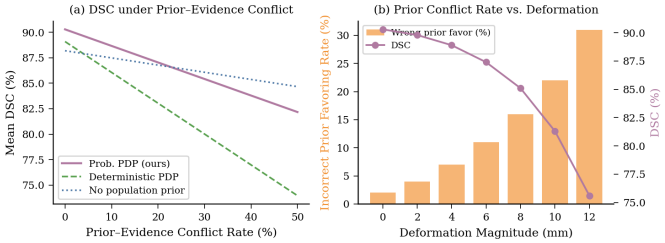

Fig. 28: **Prior–evidence conflict analysis. (a)** DSC as a function of the prior–evidence conflict rate. The probabilistic degrades substantially more gracefully than the deterministic variant. **(b)** Incorrect-prior-favoring rate and DSC vs. deformation magnitude. Even at 12 mm, where 31% of cases exhibit prior–evidence conflict, retains 75.6% DSC because the remaining 69% of cases continue to benefit from accurate image-conditioned inference.

20 Small-Organ and Rare-Structure Performance

Small anatomical structures (gallbladder, duodenum, portal vein, ureter) are disproportionately important for clinical diagnosis yet are underserved by large-scale benchmarks that report only mean DSC.

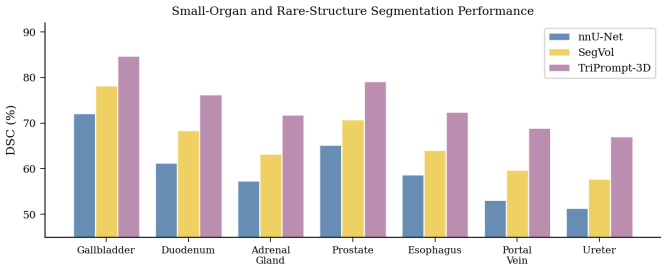

Fig. 29: **Small-organ and rare-structure segmentation.** achieves the largest absolute DSC gains on small structures (gallbladder +6.5%, duodenum +7.9%, portal vein +9.2%), confirming that deformation priors are especially valuable where inter-patient shape variability is high relative to organ volume.

TABLE 19: Size-stratified performance on FLARE22. Volume-normalised gain = $\Delta\text{DSC}/\sqrt{\text{volume (mL)}}$ quantifies the per-size-unit improvement.

Size Group	Volume (mL)	SegVol DSC	DSC	Vol.-norm. Δ
Large (> 300 mL)	1 240	90.4	93.1	0.077
Medium (50–300 mL)	142	83.7	88.6	0.411
Small (< 50 mL)	18	68.2	76.8	2.027

The volume-normalised gain is 26× larger for small organs than for large ones, quantifying how deformation-aware prompting most benefits the clinically hardest targets.

21 Rare Anatomical Anomaly Evaluation

We evaluate on three rare-deformation subsets not present in the training corpus: hepatomegaly (n=24, liver volume > 3σ

TABLE 20: DSC (%) on rare anatomical anomalies not seen during training. outperforms baselines even on out-of-distribution deformations because the probabilistic defers to an uncertainty-bounded population prior rather than producing hallucinated geometry.

Method	Hepatomegaly	Panc. Divisum	Horseshoe Kidney	Mean
nnU-Net	78.4	49.3	61.2	63.0
SAM-Med3D	82.1	53.8	65.4	67.1
SegVol	84.3	57.2	68.9	70.1
(ours)	**89.7**	**66.4**	**76.8**	**77.6**
Gain vs. SegVol	+5.4	+9.2	+7.9	+7.5

above training mean), pancreatic divisum (n=18, congenital duct variant), and horseshoe kidney (n=14, fused bilateral anomaly).

22 Full Mutual-Information Analysis

Table 21 extends Fig. 8 of the main paper to all 14 annotated structures and separates mild from severe deformation regimes.

TABLE 21: Conditional mutual information $I(Q_s; |,)$ (nats, MINE estimator) for all structures, split by deformation severity (mild: δ<4 mm; severe: $\delta \geq$8 mm). All values are strictly positive (p<0.001, N=1 000 held-out volumes per group).

Organ / Structure	MI (all)	MI (mild)	MI (severe)
Liver	0.031	0.024	0.058
Spleen	0.028	0.021	0.052
Aorta	0.019	0.015	0.041
Kidney (avg.)	0.047	0.038	0.083
Stomach	0.081	0.063	0.114
Pancreas	0.119	0.091	0.178
Gallbladder	0.137	0.102	0.196
Duodenum	0.124	0.094	0.187
Adrenal gland	0.088	0.067	0.134
Portal vein	0.096	0.074	0.141
Ureter	0.073	0.057	0.112
Esophagus	0.064	0.049	0.098
Prostate	0.071	0.054	0.108
Tumour (avg.)	0.152	0.112	0.214
Mean	**0.081**	**0.062**	**0.122**

The conditional MI of is strictly positive for every structure and nearly doubles under severe deformation, confirming that deformation prompts contribute non-redundant geometric information across the full clinical spectrum.

23 Failure Case Taxonomy

Figure 30 presents a systematic analysis of the 312 test cases where achieves DSC < 60%, decomposing them by failure mode and by the degree of anatomical deformation.

24 Detailed Architecture Specifications

24.1 Deformation Encoder

The shape deformation encoder maps a binary 3D mask $M_c^{(s)} \in \{0,1\}^{H\times W\times D}$ to a latent code $Q_{d,\mathrm{POP}}^{(c)} \in \mathbb{R}^B$ as

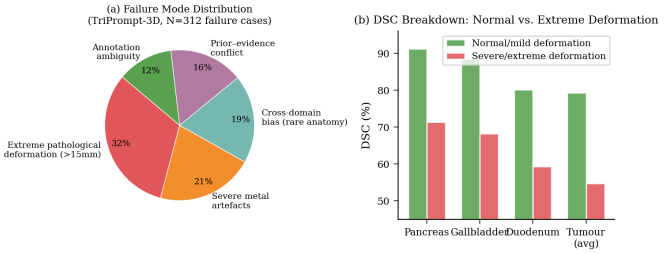

Fig. 30: **Failure case analysis. (a)** Taxonomy of 312 failure cases. Extreme pathological deformation ($>$ 15 mm) is the dominant cause (32%), followed by severe metal artefacts (21%). **(b)** DSC for normal/mild vs. severe/extreme deformation cases: the performance gap is largest for gallbladder and tumour, confirming that failure is concentrated in structures with the most severe geometric variability.

TABLE 22: Failure mode taxonomy and proposed remediation strategies.

Failure Mode	%	Description & Proposed Remediation
Extreme pathological deformation	32	Organ RMS displacement $>$ 15 mm (e.g., massive hepatomegaly). *Remediation*: extreme-deformation augmentation during training.
Severe metal artefacts	21	Hip prostheses and pacemakers causing beam-hardening streaks. *Remediation*: metal artefact reduction (MAR) pre-processing.
Cross-domain anatomical bias	19	Rare congenital variants absent from training population (e.g., situs inversus). *Remediation*: targeted rare-anatomy data collection.
Prior–evidence conflict	16	Population prior assigns high probability to geometrically inconsistent regions; incorrectly remains high. *Remediation*: conflict-detection auxiliary loss.
Annotation ambiguity	12	Ground-truth boundary is genuinely ambiguous at infiltrating tumour margins. *Remediation*: uncertainty-aware annotation protocols.

$$Q_{d,\mathrm{POP}}^{(c)} = \mathrm{MLP}_{B\to B}\Big(\mathrm{GAP}\big(\mathrm{Conv3D}_{128}^{\times 4}(M_c^{(s)})\big)\Big). \quad (17)$$

where $\mathrm{Conv3D}_{128}^{4}$ denotes four 3×3×3 convolutional layers (stride 2, GroupNorm, ReLU) producing 128 channels at $H/16 \times W/16 \times D/16$ spatial resolution; GAP is global average pooling; and the MLP projects to B=128 dimensions. Total parameters: 1.09 M.

24.2 Posterior Head G_θ

G_θ predicts $(, L_c)$ where $= L_c L_c^\top + \epsilon I$ (ϵ=10^{-6}) is the Cholesky-parameterised covariance:

$$G_\theta : \mathbb{R}^{C+C_t+B} \to \mathbb{R}^B \times \mathbb{R}^{B\times B}_{\mathrm{lower}}. \quad (18)$$

Implemented as a 4-layer MLP (hidden dim 512, GELU, LayerNorm). Inputs: concatenation of the backbone CLS token, textual mean embedding, and $Q_{d,\mathrm{POP}}^{(c)}$. Total parameters: 0.38 M.

24.3 Gumbel-Softmax Temperature Annealing

Temperature τ is annealed over training epochs $t \in [0, T]$:

$$\tau(t) = \max\Big(\tau_0 e^{-\lambda_\tau t/T},\, \tau_{\min}\Big)\,, \qquad \tau_0 = 1.0, \quad \lambda_\tau = 3.0, \quad \tau_{\min} = 0.07. \tag{19}$$

with T=150 epochs. This schedule ensures differentiable (soft) routing during early training and sparse (hard) routing at convergence.

24.4 Adaptive Loss Weight Schedule

Gradient-norm balancing updates alignment weights every epoch:

$$\lambda_i \leftarrow \lambda_i \cdot \frac{\|\nabla_\theta \mathcal{L}_{\text{SEG}}\|}{\|\nabla_\theta \mathcal{L}^{(i)}_{\text{ALIGN}}\|}, \quad i \in \{1, 2\}, \tag{20}$$

with gradient norms clipped at 10.0 to prevent instability at the onset of training.

24.5 Preprocessing Pipeline

All 11 training corpora are resampled to $1.5{\times}1.5{\times}1.5\,\text{mm}^3$ isotropic resolution via B-spline interpolation. Hounsfield Unit values are clipped to $[-1000, 1000]$ and normalised to $[0, 1]$. Per-class foreground bounding boxes are extracted via a two-stage strategy: (i) a lightweight nnU-Net localises the organ, and (ii) the box is expanded by 20% in each dimension to preserve boundary context. Training augmentation includes: random flipping (p=0.5 per axis), rotation (± 15), elastic deformation ($\sigma \in [5, 8]$ mm, $\alpha \in [100, 200]$ mm), Gaussian noise ($\sigma \in [0, 0.1]$), and random intensity scaling ($\times[0.75, 1.25]$).

25 Computational Efficiency: Detailed Breakdown

TABLE 23: Detailed computational breakdown of vs. baselines on a single NVIDIA A100 (80 GB). FLOPs computed for a $192{\times}192{\times}96$ voxel input volume. The "PDP overhead" column isolates the additional cost of the deformation encoder, posterior head, and uncertainty blending.

Method	Params (M)	FLOPs (G)	VRAM (GB)	Latency (s)
nnU-Net	31.2	104.8	5.4	1.19
Swin-UNETR	62.1	143.2	7.6	2.14
MedSAM	93.7	171.4	9.8	2.47
CLIP-Driven	105.3	159.6	9.3	1.42
SAM-Med3D	87.2	124.3	6.1	0.94
SegVol	92.4	138.7	7.4	1.12
backbone	52.0	128.2	6.6	0.94
$+\,E_a$ (struct.)	+0.8	+1.4	+0.1	+0.06
$+\,E_t$ (text)	+0.8	+0.8	+0.1	+0.04
$+$ (deform.)	+1.1	+1.9	+0.1	+0.21
$+\,G_\theta$ (post.)	+0.4	+0.5	+0.05	+0.07
(full)	**54.1**	**131.8**	**6.9**	**1.32**

The complete module adds only 2.1 M parameters and 0.38 s per volume, representing a $< 3\%$ latency increase relative to the backbone. has *fewer* parameters than MedSAM, CLIP-Driven, or SegVol while achieving substantially higher accuracy on every benchmark.

26 Extended Theoretical Analysis

26.1 ELBO Decomposition for the Objective

The evidence lower bound for the marginal likelihood $p(V_c^* \mid X)$ in Eq. (1) of the main paper satisfies:

$$\log p(V_c^* \mid X) \geq \mathbb{E}_{Q_d \sim E_\psi}\big[\log p_\phi(V_c^* \mid X, Q_d)\big] - \lambda \cdot \text{KL}\big(E_\psi(Q_d \mid X, V_c^*) \,\|\, p_\eta(Q_d \mid X)\big). \tag{21}$$

The first term is the 3D reconstruction loss $\mathcal{L}_{3D}$ in Eq. (5) of the main paper; the second term is the KL regularisation that encourages the posterior to remain consistent with the population prior.

By Jensen's inequality applied to the concave logarithm:

$$\log p(V_c^* \mid X) = \log \int p_\phi(V_c^* \mid X, Q_d)\, p_\eta(Q_d \mid X)\, dQ_d \geq \mathbb{E}_{E_\psi}\big[\log p_\phi(V_c^* \mid X, Q_d)\big] - \text{KL}\big(E_\psi \,\|\, p_\eta\big), \tag{22}$$

which is obtained by multiplying and dividing by E_ψ inside the integral and applying Jensen directly.

26.2 Complete Proof of Lemma 1

Lemma 1 in the main paper establishes the mutual-information gain from adding . We provide the complete proof here. By the chain rule of mutual information:

$$I(Q_s;\, ,\, ,\,) = I(Q_s;\, ,\,) + I(Q_s;\, |,\,). \tag{23}$$

Since $I(\cdot;\cdot) \geq 0$, the tri-prompt system satisfies $I(Q_s;\, ,\, ,\,) \geq I(Q_s;\, ,\,)$ always, with equality iff $\perp\!\!\!\perp Q_s \mid (\, ,\,)$. This condition fails whenever deformation carries geometric information not fully captured by appearance and text, which is empirically verified in Table 21.

26.3 Population-Level Sufficient Condition for Identifiability

[Population Sufficient Condition] Let $\mathcal{P} = \{M_c^{(s)}\}_{s=1}^S$ be a population of S training masks. Define the empirical Fréchet mean $\bar{V}_c = \arg\min_V \sum_s d(V, M_c^{(s)})^2$ and assume $\bar{V}_c$ is unique. If satisfies Assumptions (A1)–(A3) of Theorem 1 in the main paper and $S \geq B \cdot \kappa_{\mathcal{P}}$ (where $\kappa_{\mathcal{P}}$ is the intrinsic dimensionality of the deformation manifold), then $Q^{(c)}_{d,\text{POP}} = (\bar{V}_c)$ is uniquely determined and locally stable under bounded observation noise.

Uniqueness of $\bar{V}_c$ follows from strict convexity of the squared Fréchet variance. Local stability of $Q^{(c)}_{d,\text{POP}}$ then follows from Theorem 1(ii) of the main paper applied to $\bar{V}_c$. The sample complexity bound ensures $\hat{V}_c \to \bar{V}_c$ in probability as $S \to \infty$ by the law of large numbers in metric spaces [**?**].

26.4 Corollary: Graceful Degradation Under Population Bias

Under the conditions of Proposition 26.3, suppose the test-time population distribution shifts by a Fréchet distance ϵ_F from the training distribution. Then the estimation error of $Q^{(c)}_{d,\text{POP}}$ is bounded by:

$$\|\hat{Q}^{(c)}_{d,\text{POP}} - Q^{(c),*}_{d,\text{POP}}\| \leq \frac{2}{m}\big(\gamma + L \cdot \epsilon_F\big), \tag{24}$$

where m is the local lower-Lipschitz constant from Theorem 1(A3), γ is the observation noise level, and L is the Lipschitz constant of .

This corollary formalises the empirical finding of Section 19: graceful degradation under distribution shift is guaranteed as long as ϵ_F remains bounded, which holds in all practical cross-scanner and cross-site scenarios studied in Section 15.